Certificate Complexity Can Exceed Intrinsic Support in Quantum Compilation

Author information to be supplied before submission

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Abstract

A finite-support certificate can be exact for its proof language and still be loose as a description of the compiler it certifies. We formalize this separation with an alphabet-restricted zero-sum invariant. For a finite abelian signature group H and allowed alphabet $A \subseteq H$, let $\mathrm{zsf}(H; A)$ be the largest length of a zero-sum-free word over A. In the deletion language whose only shortening step removes a nonempty proper zero-sum subword from a nonzero-total word, the exact uniform terminal complexity is $\mathrm{zsf}(H; A)$. Every longer word is reducible, while a longest zero-sum-free word is a matching terminal witness. A production compiler inherits the matching lower bound only if that witness is realizable and no additional rule can reduce it.

Two quantum-compilation families give opposite controls. In one-Tag R6M, the production alphabet realizes $\mathbb{F}_2^2$; the deletion certificate and the independently proved intrinsic support are both two. In R6I, the production alphabets realize basis obstructions in $\mathbb{F}_2^5$, so the rank-only certificate is exactly five, whereas an independent whole-system Tag-relocation theorem gives intrinsic support exactly one. For t registered independent components, the corresponding budgets are $5t$ and t. A direct support enumerator therefore has search-volume ratio $\Theta(n^{4t})$ under the stated fixed-budget model. This is not an unrestricted proof or algorithm lower bound. It shows that a support number belongs to a compiler only after an intrinsic lower witness; otherwise it belongs to a named normalization or certificate language.

Keywords: quantum compilation; certificate complexity; support normalization; zero-sum deletion; exact optimization

1. Introduction

Support bounds turn unbounded exact compiler searches into finite ones. Their numerical value can nevertheless answer three different questions:

1. What support does the compiler intrinsically require?
2. What support can a particular normalization attain?
3. What support can a restricted proof language certify?

These quantities coincide in some families and differ sharply in others. The alphabet-Davenport normal-form theorem supplies a general upper certificate. Here we identify the exact complexity of its deletion language and compare it with independently established compiler support.

The distinction is operational. If a search implementation enumerates all supports up to a certificate ceiling, a loose proof language can impose a large, avoidable search volume even when the compiler has a much smaller normal form. Conversely, a small normal form is not intrinsic until a lower witness rules out smaller support.

Our contributions are:

1. an exact terminal-complexity theorem for zero-sum deletion;
2. a realization criterion separating abstract terminal words from production compiler states;
3. a tight R6M control with certificate and intrinsic support two;
4. a strict R6I separation between exact certificate complexity five and intrinsic support one; and
5. a registered product construction with additive gap $4t$ and a precisely scoped direct-enumeration consequence.

2. Three support quantities

Fix a compiler family $\mathcal{F}$ and objective C.

2.1 Intrinsic support

The intrinsic support $\kappa(\mathcal{F}, C)$ is the least k such that every instance has an exact optimum of support at most k, together with an independent witness showing that support $k-1$ cannot always suffice.

2.2 Normalization ceiling

A transformation N may show that every feasible configuration has a no-more-expensive representative of support at most B_N. Without a matching compiler lower witness, B_N belongs to the triple $(\mathcal{F}, C, N)$, not intrinsically to $(\mathcal{F}, C)$.

2.3 Certificate complexity

A proof language P restricts visible information and legal inference. Write $\beta_P(\mathcal{F}, C)$ for the least uniform support budget that P can certify on its production scope. Soundness gives

$$\kappa(\mathcal{F}, C) \leq \beta_P(\mathcal{F}, C),$$

but equality requires additional mathematics.

Quantity	Owned by	Lower-witness requirement
Intrinsic support κ	compiler family and objective	compiler obstruction
Normalization ceiling B_N	compiler, objective, and transformation	only for intrinsic interpretation
Certificate complexity β_P	compiler scope and proof language	terminal witness realizable in production

3. Exact zero-sum deletion complexity

Let H be a finite abelian group and $A \subseteq H$. A certificate word $v_1 \cdots v_w$ has nonzero total. The only legal shortening removes a nonempty proper subword whose sum is zero. Define

$$\mathrm{zsf}(H; A) = \max\{|W| : W \text{ is zero-sum-free over } A\}.$$

Theorem 1 (exact abstract certificate complexity). The maximum terminal length among all nonzero-total words over A is exactly $\mathrm{zsf}(H; A)$.

Proof. A word longer than $\mathrm{zsf}(H; A)$ contains a nonempty zero-sum subword. Its nonzero total prevents that subword from being the whole word, so a legal deletion exists. Conversely, a longest zero-sum-free word has nonzero total and admits no legal deletion. □

Corollary 2 (production realization). If every production word lies over A, its certificate ceiling is at most $\mathrm{zsf}(H; A)$. The ceiling is exact only if production realizes a longest zero-sum-free word and the certificate language has no other rule that reduces that state.

For $H = \mathbb{F}_2^d$, $\mathrm{zsf}(H; A) \le d$; an alphabet containing a basis has equality.

4. Tight control: R6M

A one-Tag R6M frame coordinate carries partner and Tag bits. The production alphabet realizes a basis of $\mathbb{F}_2^2$, giving

$$\beta_{\mathrm{rank}}(\mathrm{R6M}) = 2.$$

An independent all-size normalization proves support at most two, while an exact support-one obstruction proves necessity. Therefore

$$\kappa_{\mathrm{R6M}} = 2 = \beta_{\mathrm{rank}}(\mathrm{R6M}).$$

This control shows that the certificate is not inherently loose. Tightness depends on whether its terminal obstruction is also an intrinsic compiler obstruction.

5. Strict separation: R6I

R6I has two rank-two dependent-triple blocks with a shared two-bit Tag. Its two production block-deletion alphabets span five-dimensional binary quotients and realize basis words. Theorem 1 and production realization therefore give

$$\beta_{\mathrm{rank}}(\mathrm{R6I}) = 5.$$

A stronger transformation goes beyond rank-only deletion. It localizes each block to an anticommuting core and then relocates and reconstructs the shared Tag at the whole-system level. The all-size theorem proves support at most one, and support zero is infeasible. Hence

$$\kappa_{\mathrm{R6I}} = 1, \qquad \beta_{\mathrm{rank}}(\mathrm{R6I}) - \kappa_{\mathrm{R6I}} = 4.$$

There is no contradiction: a basis word blocks zero-XOR deletion while the successful proof changes auxiliary Tag structure that the rank-only language holds fixed.

6. Registered product amplification

Let $\mathrm{R6I}^{\times t}$ be the registered product of t components on disjoint coordinates, with additive support and no cross-component move.

Theorem 3. For every $t \geq 1$,

$$\beta_{\mathrm{rank}}(\mathrm{R6I}^{\times t}) = 5t, \qquad \kappa(\mathrm{R6I}^{\times t}) = t.$$

Proof. Componentwise upper bounds add. A realized basis obstruction in each component gives a certificate lower bound of five per component. Support zero is infeasible in every component, and the support-one normalization acts independently. Thus both lower and upper intrinsic bounds equal one per component. □

The additive gap is $4t$. This construction amplifies one mechanism; it is not evidence for a second, independent source of separation.

6.1 Consequence for direct support enumeration

For fixed local alphabet and fixed budget B, an enumerator that explicitly visits all coordinate supports of size at most B has leading growth $\Theta(n^B)$. Under that declared model, the registered product yields

$$\Theta(n^{5t}) \quad \text{versus} \quad \Theta(n^t),$$

and therefore ratio $\Theta(n^{4t})$. The statement is not a complexity-class lower bound and does not constrain algorithms that avoid direct support enumeration.

7. Relation to prior work

Sparse integer optimization already studies support bounds and lower constructions [1]. Classical and modern zero-sum theory owns Davenport constants, restricted alphabets, weighted variants, basis obstructions, and direct sums [2,3]. Proof-complexity and formal-methods research already distinguishes object difficulty from lower bounds internal to a proof language.

The residual contribution is the exact production separation in quantum compilation. The same algebraic certificate is tight for R6M, loose by a factor of five for R6I, and additively unbounded under registered products. The R6I normalization also identifies the missing proof operation - whole-system Tag reconstruction - instead of merely reporting a smaller number.

8. Reproducibility and limitations

Finite-group controls, binary basis obstructions, production alphabets, and product formulas have been checked with an independent deterministic verifier. Theorems 1 and 3 supply all-size authority.

The exact abstract theorem concerns only the stated deletion language. A production lower bound requires realization of its terminal witness. We prove no lower bound for every local, syndrome-preserving, or unrestricted proof system. The registered product forbids cross-component transformations by definition. The enumeration ratio belongs only to the direct support model. Finally, structural support does not imply a hardware speedup or any physical quantum-resource advantage.

9. Conclusion

A certificate can be sound, useful, and internally exact while measuring the wrong layer for an intrinsic interpretation. The alphabet-restricted invariant identifies the expressive ceiling of zero-sum deletion. R6M shows that this ceiling can coincide with intrinsic support. R6I shows that a whole-system transformation can cross it sharply.

The reporting rule is therefore simple: call a support number intrinsic only after an independent compiler lower witness. Otherwise identify the normalization or proof language that owns it.

Data and code availability

Deterministic verification scripts for the finite-group controls, production alphabets, basis obstructions, and product formulas accompany the submission source. Those computations corroborate the finite instances; Theorems 1 and 3 carry the all-size claims. A permanent archival identifier must be inserted before final upload.

References

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